



Polar regions

The sea in the Arctic and Antarctic is so cold, fish are in danger of freezing. Above the water it is even colder, and no large, cold-blooded animals exist. Warm-blooded animals—those able to retain body heat—predominate. Polar mammals often have two layers of fur: an underlayer of soft hairs that trap air warmed by the animal's body close to the skin, and an outer coat of coarse hairs that keeps out the fiercest gales.

Polar bear

This arctic mammal has a bulky, rounded body surrounded by fat and fur that keep it warm.

● Natural antifreeze

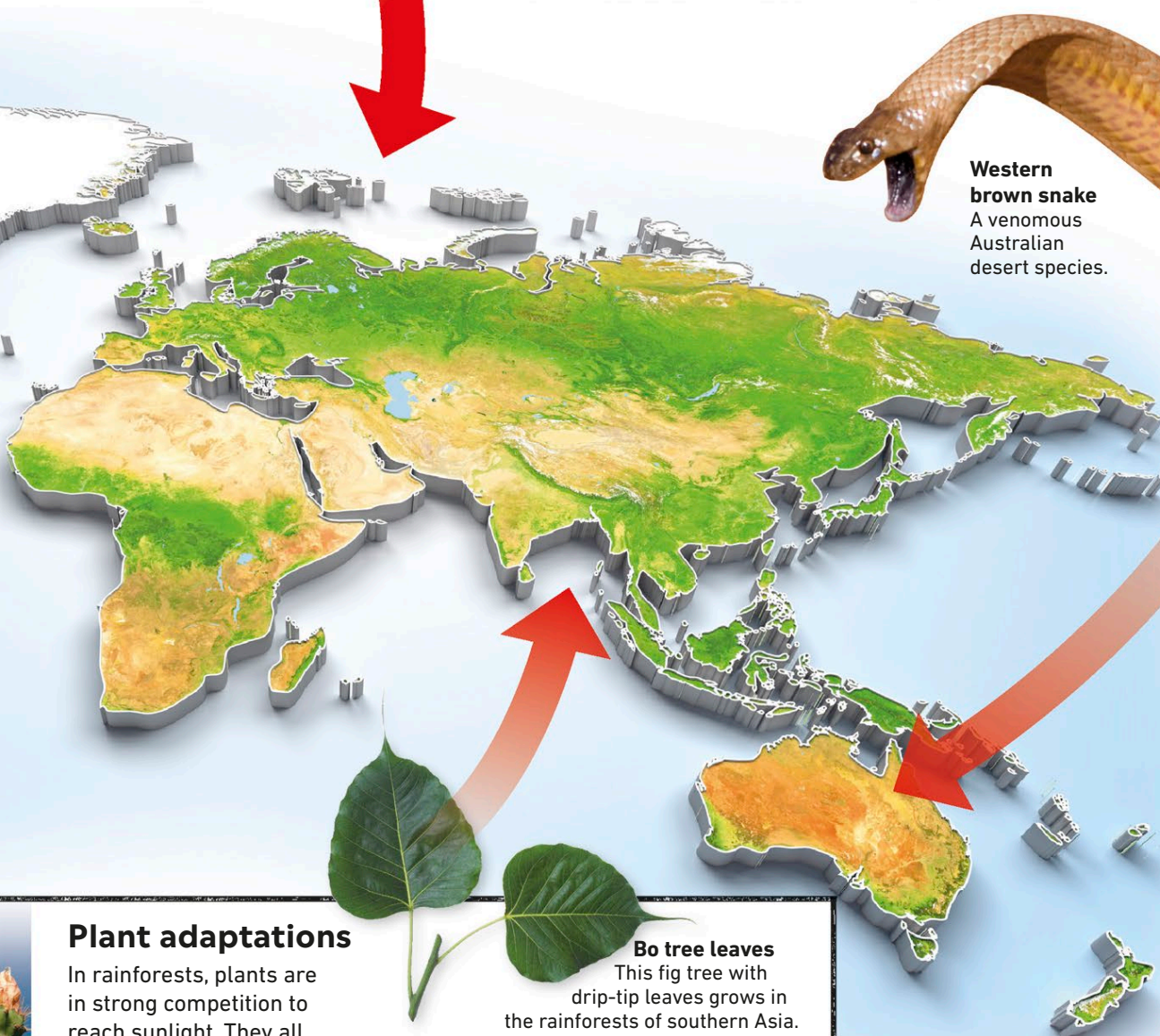
Most polar fish have a chemical in their blood that prevents ice crystals from forming in the body.

● Small extremities

Polar bears and Arctic foxes have small, rounded ears and muzzles that reduce heat loss.

● Legs and feet

Some animals have long legs that wade through snow or broad feet that act like snowshoes.



Western brown snake

A venomous Australian desert species.



Desert regions

The driest parts of the world challenge plants and animals, and desert wildlife is not as abundant as in wetter regions. Desert life-forms must get enough water—and keep what they have. Some desert animals get all the water they need from their food.

● Nocturnal lifestyle

Many animals are active only at night. Gerbils and jerboas retreat into daytime burrows to stay cool.

● Large extremities

Fennec foxes have huge ears that radiate heat away from the body.

● Drinking dew

Insects and lizards drink dewdrops. Larger desert animals that feed at dawn take in dew as they eat plants.

Plant adaptations

In rainforests, plants are in strong competition to reach sunlight. They all grow as fast as possible whenever there is an opening allowing in the sun. In deserts, plants get plenty of light, but they struggle to get enough water from the soil.

Rainforest plants

To reach the sun, many rainforest plants are specialist climbers, and others are epiphytes, which grow on top of other plants. Many rainforest leaves taper to a long point, a "drip tip," to help excess rainwater run off.

Bo tree leaves

This fig tree with drip-tip leaves grows in the rainforests of southern Asia.

